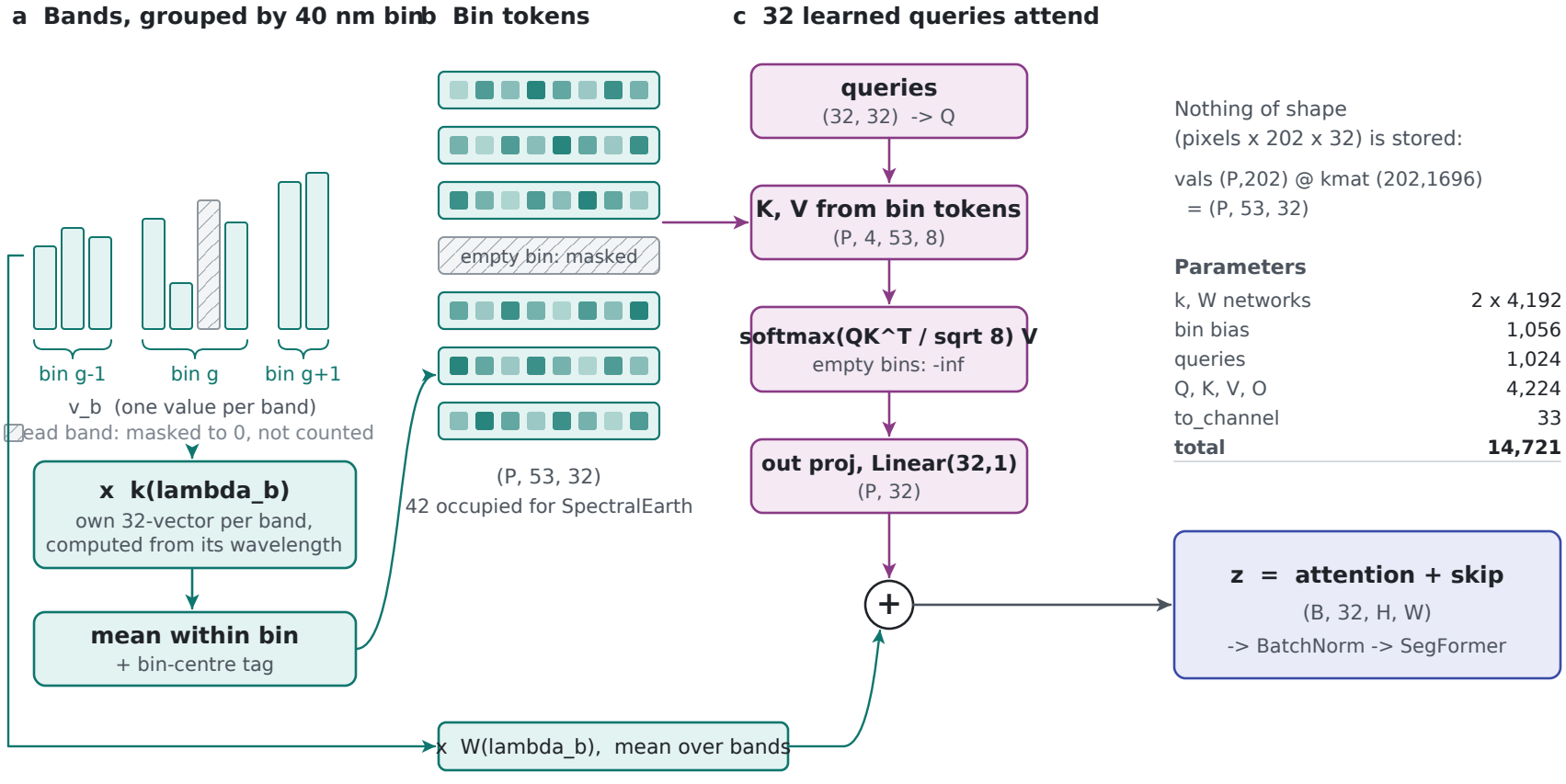
a Bands, grouped by 40 nm bin
bin g-1
bin g
bin g+1
v_b (one value per band)
ead band: masked to 0, not counted
x k(lambda_b)
own 32-vector per band,
computed from its wavelength
mean within bin
+ bin-centre tag
b Bin tokens
empty bin: masked
(P, 53, 32)
42 occupied for SpectralEarth
c 32 learned queries attend
queries
(32, 32) -> Q
K, V from bin tokens
(P, 4, 53, 8)
softmax(QK^T / sqrt 8) V
empty bins: -inf
out proj, Linear(32,1)
(P, 32)
x W(lambda_b), mean over bands
z = attention + skip
(B, 32, H, W)
-> BatchNorm -> SegFormer
Nothing of shape
(pixels x 202 x 32) is stored:
vals (P,202) @ kmat (202,1696)
= (P, 53, 32)
Parameters
k, W networks 2 x 4,192
bin bias 1,056
queries 1,024
Q, K, V, O 4,224
to_channel 33
total 14,721
d Skip path: every valid band reaches the 32 channels directly